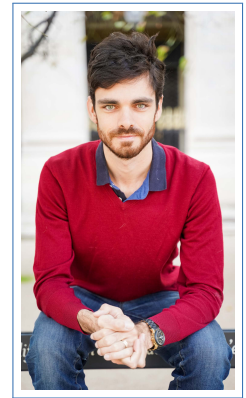


Vincent Leclère

*Researcher Hab. at École des Ponts
in Stochastic Optimization
IPEF*

42 Avenue de l'Arche
92400 Courbevoie
✉ vincent.leclere@enpc.fr
orcid:0000-0002-5757-6655



Research interests

Theory Multistage Stochastic Programming, Data-driven optimization, Decision focused learning, Decomposition methods, Dynamic programming, Convex optimization

Applications Energy, SmartGrid, Microgrid, Supply chain, Transportation, Machine learning.

Education

- 2023 **Habilitation in Mathematics (HDR)**, *UGE*, Champs sur Marne, France
Exact methods and applications of optimization under uncertainty.
- 2014 - 2015 **Post-Doctorate in Operation Research**, Berkeley, California
Robust Sketching with Laurent El Ghaoui.
- 2011 - 2014 **PhD. in Stochastic Optimization**, *ENPC*, France
Contributions to Decomposition Method in Stochastic Optimization. With Optimisation & Systèmes team of CERMICS, with M. De Lara and P. Carpentier as advisors.
- 2009 - 2011 **Master of Optimization and Game Theory**, *UPMC*, France, with very high honors
Specialization in Optimization.
- 2009 - 2011 **Master in Financial Mathematics**, *UMLV*, France, with very high honors
Stochastic processes and numerical methods.
- 2010 - 2011 **Cursus Ingénieur du Corps des Ponts, des Eaux et des Forêts**, *ENPC*
- 2006 - 2010 **Ingénieur Polytechnicien Program**, *École Polytechnique*
Applied math specialization.
- 2004 - 2006 **Preparatory school**, *Lycée Sainte Geneviève*, Versailles
Maths, Physics and chemistry.

Publications

Preprint

- 2025 **Primal-dual algorithm for contextual stochastic combinatorial optimization**, *L. Bouvier, T. Prunet, V. Leclère, A. Parmentier*
arXiv:2505.04757
- 2023 **Duality of upper bounds in stochastic dynamic programming**, *BFP Da Costa, V. Leclère*
Optimization Online preprint

Published work

- 2027 **Distributionally robust unit commitment models: Theory and numerical results**, *M. Azéma, V. Leclère, W. van Ackooij*
Forthcoming in Electric Power Systems Research, 263, 113754.
- 2026 **Stronger cuts for Benders' decomposition for stochastic Unit Commitment Problems based on interval variables**, *M. Azéma, V. Leclère, W. van Ackooij*
Accepted/in press in Mathematical Programming Computation.
- 2025 **Strategic behavior of risk-averse agents under stochastic market clearing**, *V. Leclère, A. Philpott*
Published in Operations Research Letters, 63, 107365.
- 2025 **Ten challenges for mathematical modeling of the green-energy transition**, *E. Anderson et al.*
Published in Current Sustainable/Renewable Energy Reports, 12, 26.
- 2025 **Fairness by design in shared-energy allocation problems**, *Z. Fournier, V. Leclère, P. Pinson*
Published in Computational Management Science, 22(2), 11.
- 2025 **Joint Production and Energy Supply Planning of an Industrial Microgrid**, *Z. Fournier, D. Grosso, V. Leclère*
Published in Energy Systems, 16(4), 1461–1493.
- 2024 **Policy with guaranteed risk-adjusted performance for multistage stochastic linear problems**, *Lucas Mérabet, BFP Da Costa, V. Leclère*
Computational Management Science
- 2024 **Exact quantization of multistage stochastic linear problems**, *M. Forcier, S. Gaubert, V. Leclère*
Published in SIAM journal on Optimization. *Best student paper award at CMS-ECSO 2022.*
- 2023 **Robust limit analysis theory for computing worst-case limit loads under uncertainties**, *J. Bleyer, V. Leclère*
Book Chapter in Direct Methods for Limit State of Materials and Structures: Advanced Computational Algorithms and Material Modelling
- 2023 **Convergence of trajectory following dynamic programming algorithms for multistage stochastic problems without finite support assumptions**, *M. Forcier, V. Leclère*
Published in Journal of Convex Analysis.
- 2023 **Dual sddp for risk-averse multistage stochastic programs**, *BFP. Da Costa, V. Leclère*
Published in Operations Research Letters.

- 2022 **Generalized adaptive partition-based method for two- stage stochastic linear programs: Geometric oracle and analysis**, *M. Forcier, V. Leclère*
Published in Operations Research Letters. *2022 ORL Best paper award.*
- 2020 **Integer Programming on the Junction Tree Polytope for Influence Diagrams**, *A. Parmentier, V. Cohen, V. Leclère, G. Obozinski, J. Salmon*
Published in INFORMS Journal on Optimization.
- 2020 **Exact converging bounds for Stochastic Dual Dynamic Programming via Fenchel duality**, *V. Leclère, P. Carpentier, J-Ph. Chancelier, A. Lenoir, F. Pacaud*
Published in SIAM Journal on Optimization
- 2019 **Epiconvergence of relaxed stochastic optimization problems**, *V. Leclère*
Published in Operations Research Letters.
- 2018 **On risk averse competitive equilibrium Optimization**, *H. Gérard, V. Leclère, A. Philpott*
Published in Operations Research Letters.
- 2018 **Stochastic decomposition applied to large-scale hydro valleys management** , *P. Carpentier, J-Ph. Chancelier, F. Pacaud, V. Leclère*
Published in European Journal of Operation Research
- 2016 **Building up Time-Consistency for Risk Measures and Dynamic Optimization**, *M. De Lara, V. Leclère*
Published in European Journal of Operation Research.
- 2015 **On the convergence of decomposition methods for multi-stage stochastic convex programs**, *P. Girardeau, V. Leclère, A. Philpott*
Published in Mathematics of Operation Research.
- 2013 **Priority option: the value of being a leader**, *M. Grasselli, V. Leclère, M. Ludkovski*
Published in International Journal of Theoretical and Applied Finance.

Peer-reviewed proceedings

- 2025 **Decision Focused Scenario Generation for Contextual Two-Stage Stochastic Linear Programming**, *J. Hornewall, S. Delannoy-Pavy, T. Homem-de-Mello, V. Leclère*
NeurIPS 2025 MLxOR workshop. OpenReview.
- 2018 **A stochastic multi-item lot-sizing problem with bounded number of setups**, *E. De Saint Germain, V. Leclère, F. Meunier*
ICORES 2017 Proceeding
- 2017 **Efficient Smoothed Concomitant Lasso Estimation for High Dimensional Regression**, *E. Ndiaye, O. Fercoq, A. Gramfort, V. Leclère, J. Salmon*
Published in J. Phys.: Conference Series.

Young researchers advising

PhD Students

- 2026 - ... **Alexis David**, *CIFRE funding with Schneider Electric*, Decision Focused Learning for Energy Management
- 2024 - ... **Solène Delannoy-Pavy**, *CIFRE funding with RTE*, Predictive maintenance of offshore wind farm grid connection substations
In collaboration with Axel Parmentier

- 2024 - ... **Mathis Azéma**, *DGA funding*, Distributionally robust unit commitment
In collaboration with Wim Van Ackooij
- 2023 - ... **Jonathan Hornewall**, *Ecole des Ponts and IPP's funding*, Polyhedral approaches for multiscale stochastic optimization
In collaboration with Stéphane Gaubert
- 2022 - ... **Vitor Luiz Pinto de Pina Ferreira**, *In partnership with TotalEnergies*
- 2022 - 2025 **Zoé Fournier**, *In partnership with Metron Energy*
- 2019 - 2022 **Maël Forcier**, *Ministère de la transition écologique's funding*
In collaboration with Stéphane Gaubert, supervised by Jean-Philippe Chancelier
- 2015 - 2018 **Étienne de Saint Germain**, *In partnership with Argon Consulting*
In collaboration with Frédéric Meunier

Postdoctorates

- 2021 - 2023 **Carlos Moreno**, *Chair Supply chain of Tomorrow's funding*
- 2018 - 2019 **Regan Baucke**, *DIM Math'Innov's funding*
Supervised by Jean-Philippe Chancelier

Grants and industrial contracts

Current contracts

- 2022 - 2025 **Optimization of an isolated hybrid system (PI)**, *TotalEnergies*,
Industrial contract funding for a PhD thesis aiming at optimizing the energy production plan of an isolated system with thermal production, renewable energies and energy storage.
- 2021 - 2024 **Optimization of a system coupling industrial production, renewable energy and energy storage (PI)**, *Metron Energy*,
CIFRE funding for a PhD thesis aiming at optimizing the coupled production and energy procurement short-term plan of an industrial micro-grid.

Past contracts

- 2021 - 2023 **Mathematical model and solution methods for resilient and green supply chain design (PI)**, *Chair Supply Chain of Tomorrow*,
Industrial contract with a consortium of companies, aiming at proposing a multi-objective model for supply chain design. Funding an 18 months post-doc position.
- 2021 - 2023 **Reverse logistic inventory routing (co-PI)**, *Renault*,
Industrial contract tackling a split-delivery multi-item large scale inventory routing problem. Focus on the strategical problem to select regular routes at a discounted price.
- 2019 - 2023 **Two-scale optimization problems (PI)**, *Programme Gaspard Monge pour l'Optimisation et la recherche opérationnelle - EDF*,
- 2018 - 2019 **Exact bounds for stochastic optimization (co-PI)**, *DIM-MathInnov*,
Institutional funding covering a one-year post-doctorate position.
- 2015 - 2018 **Balancing cost and flexibility in supply chain (co-PI)**, *Argon Consulting*,
CIFRE funding for a PhD thesis aiming at balancing cost and flexibility for tactical decisions (lot-sizing) and for strategical decision (sourcing strategy).
- 2016 - 2017 **Equilibrium and games in energy (PI)**, *Programme Gaspard Monge pour l'Optimisation et la recherche opérationnelle - EDF*,
- 2015 - 2016 **Epi-splines for solar energy prevision (PI)**, *Programme Gaspard Monge pour l'Optimisation et la recherche opérationnelle - EDF*,
- 2014 - 2015 **Robust Sketching for Structured Multi-Instance Optimization with Uncertainty**, *Programme Gaspard Monge pour l'Optimisation et la recherche opérationnelle - EDF*,
- 2013 - 2016 **SunHydro Project**, Optimal management of an hydroelectric storage coupled with a renewable energy production unit
- 2012 - 2013 **Décomposition/Coordination en commande optimale stochastique - StochDec**,
Programme Gaspard Monge pour l'Optimisation et la recherche opérationnelle - EDF
- 2012 - 2014 **Optimization Methods for Smart Grid**, report for the Conseil Français de l'Énergie, french member of the World Energy Council

Contributions to the scientific community

Conference and workshop organization

- 2025 Organizer of the ICSP XVII conference. This is the main event in the field of stochastic programming (env. 300 participants).
- 2023 Organizer of the 2023 Smart Energy and Stochastic Optimization (SESO) workshop at École des Ponts
- 2023 Co-organizer of the Julia Days workshop at CNAM, Paris
- 2023 Organizer of mini-symposiums on SDDP at ICSP
- 2021 - ... Co-organizer of the decision, algorithm and geometry seminar between CERMICS and CMAP
- 2021 Organizer of a workshop on robust and stochastic optimization methods at École des Ponts
- 2020 Contribution to the organization of the 2020 SMAI-MODE conference
- 2019 Co-organizer of a winter school at CIRM (Luminy)
- 2019 Co-organizer of a workshop on Multi-Stage Stochastic Optimization for Clean Energy Transition at Banff International Research Station (Oaxaca, Mexico)
- 2018 Organizer of a stream of invited sessions on SDDP at ISMP
- 2016 Organizer of a mini-symposium on SDDP at ICSP
- 2014 - 2018 Contribution to the organization of Smart Energy and Stochastic Optimization (SESO) workshop at Ecole des Ponts

Position in scientific communities

- 2023 - ... Member of the Committee on Stochastic Programming of the Mathematical Programming Society
- 2023 - 2026 Vice President for industrial relations of the SMAI steering committee
- 2023 - ... Co directing the transversal axis *Decision and Robust Optimization*, of GDR RO
- 2018 - 2024 Elected member of the SMAI-MODE steering committee

Evaluation of peers

- 2022 ROADEF best student paper award committee member
- 2020 - 2021 Nicholson Prize committee member for best paper student
- 2012 - ... Reviewer of numerous scientific journals (notably MP, MPC, SiOPT, OR, ORL, EJOR...)

Teaching

Current teachings

- 2025 - ... **Optimization***, *3rd year course*, ENPC, (15h)
- 2025 - ... **Markov Chains and Markov Decision Processes**, *3rd year course*, ENPC, (15h)
- 2023 - 2025 **Stochastic and Robust optimization***, *5th year course*, MPRO, (24h)
- 2020 - 2025 **Convex Optimization***, *4th year course*, ENPC, (30h)

Past teachings

- 2018 - 2025 **Stochastic Optimization***, *5th year course*, University Paris-Saclay, (15h)
- 2015 - 2025 **Operation Research and Transportation***, *3rd year course*, ENPC, (15h)
- 2017 - 2020 **Data Driven Robust Optimization***, *5th year course*, ENPC, (15h)
- 2015 - 2020 **Optimization and Energy***, *3rd year course*, ENPC, (15h)
- 2017 - 2020 **Finding an optimal board game strategy***, *3rd year project*, ENPC, (10h)
- 2010 - 2020 **Optimization and control**, *4th year course*, ENPC, (10h)
- 2015 - 2018 **Introduction to Optimization**, *3rd year course*, ENPC, (12h)
- 2013 - 2017 **Stochastic Optimization**, *5th year course*, MPRO, (9h)
- 2011 - 2014 **Differentiable optimization**, *4th year course*, ENSTA, (24h)
- 2011 - 2013 **Introduction to probability**, *3rd year course*, ENSTA, (24h)
- 2011 - 2013 **Le risque dans tous ses états**, *thematic week*, ENPC, (10h)

Knowledge transfer (Industrial courses)

- 2023 **Stochastic Optimization***, *TotalEnergies*, (12h)
- 2022 **Stochastic Optimization**, *RTE*, (6h)
- 2021 **Multiobjective Optimization***, *Chair Supply Chain of Tomorrow*, (3h)
- 2021 **Stochastic Optimization***, *Metron Energy*, (6h)
- 2020 - 2022 **Stochastic and Robust Optimization***, *Air France*, (8h)
- 2020 **Stochastic Optimization**, *Total Energies*, (6h)
- 2019 **Stochastic Optimization**, *CNRS Interface Winter School*, CIRM Luminy, (15h)
- 2016 **Stochastic Optimization**, *SESO Winter School*, ENPC, (6h)
- 2015 **Stochastic and Robust Optimization**, *IRT System'X*, (6h)
- 2013 **Stochastic Optimization**, *XM-Columbia*, (6h)
- 2012 **Progressive Hedging**, *practical session of 2-week summer school*, (8h)

*: course responsibility